

Problem 6.12

If \$200 grows to \$500 over n years, what will \$700 grow to over $3n$ years? Assuming same compound annual interest rate.

Let the annual effective interest rate be i . Then

$$500 = 200(1 + i)^n.$$

So,

$$(1 + i)^n = \frac{500}{200} = 2.5.$$

Therefore,

$$(1 + i)^{3n} = ((1 + i)^n)^3 = (2.5)^3 = 15.625.$$

If X is the amount that \$700 grows to in $3n$ years, then

$$X = 700(1 + i)^{3n} = 700(15.625).$$

Hence,

$$\boxed{X = 10937.5}.$$